

Prompt Engineering and Large Language Models

Makeup Examination

Whole course, weighted towards Weeks 6–13 · 40 questions × 2.5 points = 100 points

Name: _____ Student No: _____ Date: _____

Instructions. This paper contains 40 multiple-choice questions. Select the **single best** answer for each. Each question is worth equal marks. Time allowed: 90 minutes. No notes or electronic devices.

1. Which is the single operation underlying everything a large language model produces?
 - A) Matching the question against a database of answers
 - B) Predicting the next token, repeatedly
 - C) Executing rules supplied by developers
 - D) Summarising retrieved web pages
2. Why is the model unable to signal when it is guessing?
 - A) Guessing is suppressed by safety training
 - B) The same operation runs whether the pattern is strong or weak, and confidence is not part of the output
 - C) Confidence scores exist but are hidden by the interface
 - D) Users have disabled the feature
3. A model is asked about a regulation amended after its knowledge cut-off. What is most likely?
 - A) It reports that its knowledge ends at a specific date
 - B) It answers confidently using the earlier version or something adjacent
 - C) It refuses to answer
 - D) It searches for the update automatically
4. Tokenisation explains all of the following **except**:
 - A) Difficulty counting letters within a word
 - B) Inconsistent behaviour of word-count instructions
 - C) Higher cost for Turkish text than English
 - D) The tendency to accept a false premise in a question
5. Which word is most likely to be split into several tokens?
 - A) and
 - B) patient
 - C) an uncommon Turkish word built from several suffixes
 - D) the
6. What happens when a document exceeds the context window in the most dangerous case?
 - A) The service returns an explicit error

- B) Part of the material is dropped silently and a confident answer follows
 - C) The answer is visibly cut off
 - D) The model asks for a shorter document
7. Which statement about fine-tuning for safety is accurate?
- A) It installs a rule that blocks disallowed output
 - B) It shifts probabilities, producing a tendency rather than a guarantee
 - C) It removes disallowed material from training data
 - D) It applies only to images
8. Two colleagues get different answers to an identical prompt. The best explanation is:
- A) One of them used a different model
 - B) Randomness in token selection
 - C) One prompt contained a typographical error
 - D) The context window differed in size
9. Which lever should be pulled when the output assumes the wrong reader or the wrong country?
- A) Format
 - B) Examples
 - C) Context
 - D) Temperature
10. Which is the strongest reason for asking a model to quote its supporting sentence?
- A) It shortens the answer
 - B) It converts an unverifiable claim into one you can check in seconds
 - C) It prevents hallucination
 - D) It reduces token cost
11. Why is “make it more professional” an unreliable instruction?
- A) The model does not recognise the word
 - B) It describes a feeling rather than a text, and many different outputs satisfy it
 - C) It conflicts with safety training
 - D) It requires more tokens than the alternatives
12. In the three dimensions of register, **mode** refers to:
- A) The subject vocabulary
 - B) The relationship between writer and reader
 - C) The channel — written to be read, written to be spoken, planned or spontaneous
 - D) The level of formality
13. Why does specifying a fallback string reduce invented content?
- A) It lowers the temperature
 - B) It gives the honest answer a fixed and probable form
 - C) It blocks access to training data

- D) It shortens the response
14. A summary is factually correct throughout but omits the only finding that would have changed a decision. This is:
- A) Fabrication
 - B) Displacement
 - C) Omission
 - D) Over-reading
15. An answer about local practice accurately describes the practice of a different country. This is:
- A) Displacement
 - B) Fabrication
 - C) Silent failure
 - D) Leakage
16. An assistant grounded in your documents cites a genuine sentence that does not support the claim it is attached to. This is:
- A) Answered from memory
 - B) Retrieved the wrong passage
 - C) Omission
 - D) Fabrication
17. Your source states a requirement applies “in exceptional cases”. The assistant reports it applies generally. This is:
- A) Over-reading the passage
 - B) Fabrication
 - C) Displacement
 - D) Retrieved the wrong passage
18. An automation produces a polished result from corrupted input, with no error anywhere. This is:
- A) Wrong stopping point
 - B) Silent failure
 - C) Context overflow
 - D) Direct injection
19. What is common to every failure mode covered in this course?
- A) They require long conversations
 - B) They are invisible in the output
 - C) They affect only grounded systems
 - D) They can be eliminated by better prompts
20. Which is genuine verification of a factual claim?
- A) Asking the model how confident it is
 - B) Asking the same question a second time

- C) Locating the claim in a source you trust
 - D) Asking the model to double-check its work
- 21.** A generated tool works on the inputs you tried and fails on an empty field. Which failure is this?
- A) Wrong logic with right appearance
 - B) Correct only for the cases tried
 - C) Silently discarded input
 - D) Indirect injection
- 22.** Why is a consistently wrong generated tool more dangerous than an intermittently wrong one?
- A) It uses more resources
 - B) Consistency resembles reliability, so the error is not noticed
 - C) It is harder to regenerate
 - D) It fails at higher volumes
- 23.** An agent removes which safeguard that users rarely realise they were providing?
- A) Rate limiting
 - B) Inspection of every intermediate output
 - C) Context management
 - D) Source selection
- 24.** Five steps at 95% reliability gives an end-to-end reliability closest to:
- A) 95%
 - B) 90%
 - C) 77%
 - D) 50%
- 25.** Why does the “am I finished?” decision cause difficulty for agents?
- A) It requires access to external tools
 - B) It is a judgment about matching an intention that is not in the context, so it is predicted like anything else
 - C) It consumes excessive tokens
 - D) It is disabled by default
- 26.** Which is the most robust control over an agent’s behaviour?
- A) A detailed instruction to be cautious
 - B) A request that it explain its reasoning
 - C) Connecting only the tools strictly required
 - D) Setting a low temperature
- 27.** A workflow drafts patient letters and sends them automatically. Which design rule is violated?
- A) Keep the chain short
 - B) Make each step inspectable

- C) Put the human before the irreversible action
 - D) Use the narrowest tools
- 28.** Which is the best candidate for a first AI workflow?
- A) The most intellectually demanding task in your field
 - B) A repetitive task whose answers exist in a document, where a mistake is recoverable
 - C) A task requiring access to personal records
 - D) A task nobody currently performs
- 29.** Answering “users might receive incorrect information” to the question of who is harmed is inadequate because:
- A) It is too brief
 - B) It commits to nothing and therefore changes no design decision
 - C) Users are not the only stakeholders
 - D) It omits the probability of harm
- 30.** In a well-designed workflow, most decisions concern:
- A) Model selection
 - B) Constraints — what the system must not do
 - C) Prompt phrasing
 - D) Cost optimisation
- 31.** Which does a Domain Pack **not** address?
- A) Institution-specific vocabulary
 - B) The structure professionals expect
 - C) What must never be stated without qualification
 - D) Whether the content is true
- 32.** Why is professional formatting a double-edged improvement?
- A) It consumes context
 - B) It makes incorrect content more persuasive
 - C) It slows the workflow
 - D) It reduces the refusal rate
- 33.** Why does asking for IRAC structure help with legal analysis?
- A) It is shorter than prose
 - B) Stating the rule before applying it makes a missing or wrong rule visible
 - C) Lawyers prefer lists
 - D) It reduces hallucination generally
- 34.** What makes prompt injection difficult to eliminate?
- A) Attackers are highly skilled
 - B) Instructions and data arrive through the same undifferentiated channel
 - C) Context windows are too small
 - D) Safety training is incomplete

- 35.** A colleague summarises a PDF containing hidden instructions and receives a subtly altered summary. This is:
- A) Direct injection
 - B) Indirect injection
 - C) Jailbreaking
 - D) Leakage
- 36.** Which attacker objective becomes possible only when tools are connected?
- A) Altering the content of an answer
 - B) Extracting other documents from the context
 - C) Sending, deleting or purchasing
 - D) Revealing system instructions
- 37.** Sharing a link to a conversation is riskier than sharing a screenshot because:
- A) Links can be forwarded more easily
 - B) The link carries the entire history, including material you have forgotten
 - C) Screenshots are encrypted
 - D) Links expire unpredictably
- 38.** Which statement about anonymising clinical material is correct?
- A) Removing names is sufficient
 - B) Rare details in a small population can identify a person even without names
 - C) Anonymised data may be uploaded anywhere
 - D) Anonymisation applies only to published research
- 39.** Under this course's reasoning, what makes undisclosed AI use a problem in assessed work?
- A) AI is inherently dishonest
 - B) It breaks the measurement the assessment was designed to make
 - C) It produces lower-quality work
 - D) It is difficult to detect
- 40.** Why keep one professional skill that you practise unaided?
- A) Institutional rules require it
 - B) The judgment needed to evaluate generated work is built by producing work yourself
 - C) Generated output is generally poor
 - D) It is faster in the long term